

VOJENS AP, Denmark

WMO: 061100

Lat: 55.2253N

Lon: 9.2633E

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.1	(c) -6.8	(d) -11.0	(e) 1.5	(f) -7.7	(g) -8.9	(h) 1.8	(i) -5.2	(j) 15.8	(k) 7.1	(l) 14.2	(m) 7.2	(n) 2.0	(o) 10	(p) 0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	26.7	18.9	24.7	18.5	22.8	17.4	20.1	24.3	19.1	23.1	18.1	21.7	4.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	22.0	17.7	12.7	20.8	16.8	12.0	19.8	57.7	24.2	54.3	23.0	51.2	21.7	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.3	(b) 10.9	(c) 9.6	(e) -13.2	(f) 29.1	(g) 3.6	(h) 1.7	(i) -15.7	(j) 30.3	(k) -17.8	(l) 31.3	(m) -19.8	(n) 32.3	(o) -22.4	(p) 33.6		
(a) 12.3	(b) 10.9	(c) 9.6	(e) -13.5	(f) 22.0	(g) 3.4	(h) 0.9	(i) -16.0	(j) 22.7	(k) -18.0	(l) 23.2	(m) -19.9	(n) 23.8	(o) -22.4	(p) 24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	1.1	1.3	3.4	7.4	11.2	14.0	16.5	16.5	13.5	9.3	5.1	2.2
	DBStd	6.51	3.90	3.85	3.37	3.36	3.21	2.81	2.88	2.73	2.58	3.15	3.43	4.24
	HDD10.0	1286	277	243	206	91	24	2	0	0	3	50	148	242
	HDD18.3	3628	535	476	463	327	223	134	73	71	147	280	397	501
	CDD10.0	738	0	0	1	14	61	121	202	202	107	29	2	0
	CDD18.3	37	0	0	0	0	1	3	17	15	1	0	0	0
	CDH23.3	310	0	0	0	1	7	37	153	108	4	0	0	0
	CDH26.7	45	0	0	0	0	0	4	24	17	0	0	0	0
Wind	WSAvg	4.6	5.3	5.4	5.2	4.6	4.3	4.4	4.0	3.8	4.1	4.4	4.6	4.9
Precipitation	PrecAvg	873	81	62	55	40	54	71	76	98	85	92	76	83
	PrecMax	1078	166	139	124	115	99	142	172	200	169	201	163	168
	PrecMin	594	5	14	9	3	12	2	14	37	12	41	15	27
	PrecStd	127	39	26	28	23	23	34	38	43	39	31	36	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	10.0	14.2	21.0	24.1	27.0	29.1	28.8	23.8	18.1	13.9	11.0
		MCWB	9.1	8.0	9.5	14.1	16.9	18.7	19.6	19.4	18.5	15.4	12.8	10.1
	2%	DB	8.2	8.2	11.3	17.8	21.6	24.1	27.0	26.2	21.2	16.1	12.2	9.8
		MCWB	7.6	7.5	8.2	12.1	15.4	18.0	19.1	18.8	16.8	14.2	11.2	9.3
	5%	DB	7.2	7.2	9.3	15.4	19.4	21.8	25.0	24.1	19.4	14.9	11.0	8.7
		MCWB	6.6	6.5	7.2	10.8	14.1	16.6	18.6	18.1	15.8	13.4	10.0	8.2
	10%	DB	6.2	6.4	8.0	13.1	17.5	19.8	22.9	22.0	18.0	13.9	9.9	7.8
		MCWB	5.7	5.7	6.7	9.8	13.2	15.5	17.5	17.5	15.1	12.6	9.3	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	8.6	10.4	15.1	18.4	20.5	21.5	21.2	19.0	15.9	13.1	10.4
		MCDB	9.4	9.1	12.6	20.2	22.9	24.5	26.3	25.2	22.4	17.3	13.7	10.8
	2%	WB	7.7	7.5	8.7	12.8	16.3	18.7	20.3	19.8	17.5	14.7	11.5	9.2
		MCDB	8.0	8.0	10.6	16.5	19.9	22.9	25.0	23.8	20.3	15.7	12.2	9.6
	5%	WB	6.7	6.7	7.7	11.3	14.9	17.3	19.2	18.9	16.5	13.7	10.3	8.1
		MCDB	7.0	7.2	8.8	14.5	18.3	20.7	23.4	22.6	18.7	14.6	10.8	8.5
	10%	WB	5.7	5.8	6.7	10.2	13.7	16.0	18.1	18.0	15.6	12.8	9.2	7.2
		MCDB	6.0	6.3	7.7	12.7	16.9	19.0	21.7	21.4	17.4	13.8	9.7	7.6
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	6.7	8.7	9.4	9.0	9.4	9.2	7.8	6.4	5.1	4.6
		MCDBR	4.3	5.0	8.9	11.8	12.6	12.2	12.6	12.0	9.8	7.1	5.0	4.3
		MCWBR	4.3	4.3	6.2	7.2	7.5	7.1	6.2	5.9	6.0	5.4	4.4	4.2
	5% WB	MCDBR	4.2	4.7	7.6	10.4	11.0	10.7	11.4	10.4	8.3	6.4	4.7	4.2
		MCWBR	4.3	4.2	5.8	6.8	7.0	6.9	6.3	5.7	5.4	5.3	4.5	4.3
Clear-Sky Solar Irradiance	taub	0.321	0.340	0.365	0.397	0.390	0.385	0.401	0.401	0.379	0.357	0.335	0.309	
	taud	2.254	2.291	2.272	2.232	2.311	2.348	2.321	2.333	2.377	2.383	2.313	2.249	
	Ebn at Noon	589	726	797	821	853	860	837	811	778	700	570	513	
	Edh at Noon	58	79	102	121	119	116	117	109	92	72	55	47	
All-Sky Solar Radiation	RadAvg	0.50	1.12	2.43	4.06	5.26	5.53	5.17	4.15	2.90	1.52	0.62	0.36	
	RadStd	0.08	0.16	0.22	0.47	0.52	0.44	0.56	0.30	0.30	0.18	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.36	N/A	N/A	N/A	N/A	N/A	N/A	-147	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-145	N/A	N/A

Nomenclature: See separate page